Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A carnot engine whose low temperature reservoir is at $7^{\circ}\mathrm{C}$ has an efficiency of 50%. It is desired to increase the efficiency to 70%. By how many degrees should the temperature of
the high temperature reservoir be increased

(A) $933\ \mathrm{K}$

(B) $380\ \mathrm{K}$

(C) $560\ \mathrm{K}$

(D) $373\ \mathrm{K}$

Q.2

A Carnot's engine, with a perfect gas acting as the working substance, is operated backwards and is employed to freeze water at $0^{\circ}\mathrm{C}$. The engine is driven by a $200\ \mathrm{W}$ electric motor. Find the amount of water frozen per minute. The outside temperature is $20^{\circ}\mathrm{C}$. Specific latent heat of fusion of $\mathrm{ice} = 80\mathrm{cal/g}$ and $\mathrm{J} = 4.2\ \mathrm{J/cal}$.

(A) 487.5 g/min

(B) 452.5 g/min

(C) 472.5 g/min

(D) 457.5 g/min

Q.3

A refrigerator is devised by operating a Carnot's heat engine in the reverse direction. If the inside temperature is $-10^{\circ}\mathrm{C}$ and that of outside is $30^{\circ}\mathrm{C}$, find its coefficient of performance.

(A) 3.4

(B) 2.5

(C) 5.9

(D) 35.52

Q.4

A heat engine receives $50\mathrm{kcal}$ of heat from the source per cycle, and operates with an efficiency of 20%. Determine t he work obtained from the engine and the heat rejected to sink, per cycle.

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(A) 34 kcal

(B) 25 kcal

(C) 40 kcal

(D) 35.52 kcal

Q.5

A motor cycle engine delivers a power of $10\ \mathrm{kW}$ by consuming petrol at the rate of $2.4\ \mathrm{kg/}$ hour. Estimate the efficiency of the engine and the rate of heat rejection by the exhaust, if the calorific value of petrol is $35.5\mathrm{MJ/kg}$.

(A) 10 kW

(B) 15.7 kW

(C) 23.7 kW

(D) 13.7 kW

Q.6

A heat engine rejects 75% of the heat supplied to the surrounding (acting as a sink) per cycle. Estimate its efficiency.

(A) 34. %

(B) 25 %

(C) 44.52 %

(D) 35.52 %

Q.7

An ideal Carnot's engine works between temperatures $127^\circ\mathrm{C}$ and $-73^\circ\mathrm{C}$. Find, which of the two processes, increasing the temperature of source by $50^\circ\mathrm{C}$, or decreasing the temperature of sink by $50^\circ\mathrm{C}$, would result in a more increase in the efficiency of the engine. The value of the greater efficiency will be -

(A) 34.2 %

(B) 33.5 %

(C) 44.52 %

(D) 62.5 %

Q.8

A Carnot's heat engine works between the temperatures $427^\circ\mathrm{C}$ and $27^\circ\mathrm{C}$. What amount of heat should it consume per second to deliver mechanical work at the rate of $1.0\ \mathrm{kW}$?

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(A) 0.3452 Kcal/s

(B) 0.417 Kcal/s

(C) 0.342 Kcal/s

(D) 0.399 Kcal/s

Q.9

A Carnot's heat engine works with an ideal monatomic gas, and an adiabatic expansion ratio 2. Determine its efficiency.

(A) 34.52 %

(B) 37 %

(C) 44.52 %

(D) 35.52 %

Q.10

A Carnot's engine operates with an efficiency of 40% with its sink at $27^{\circ}\mathrm{C}$. By what amount should the temperature of the source be increased with an aim to increase the efficiency by 10% ?

(A) 100 K

(B) 300 K

(C) 490 K

(D) 380 K

Q.11

Calculate the thermal efficiency of a Carnot's heat engine working between ice and steam point.

(A) 25.54 %

(B) 26.81 %

(C) 32.76 %

(D) 24.69 %

Q.12

Which statement is incorrect?

(A) All reversible cycles have the same efficiency

(B) Reversible cycle has a higher efficiency than an irreversible cycle

(C) Carnot cycle is a reversible cycle

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(D) Carnot cycle has the maximum efficiency of all cycles

Q.13

Even Carnot engine does not have 100% efficiency because we cannot

(A) Prevent heat loss due to radiation

(B) find an ideal source of heat

(C) attain absolute zero temperature

(D) eliminate friction

Q.14

A diatomic ideal gas is used in Carnot engine as the working substance. If during the adiabatic expansion part of the cycle, volume of the gas increases from V to $32V$ the efficiency of the engine is

(A) 0.5

(B) 0.75

(C) 0.99

(D) 0.25

Q.15

A Carnot engine, having an efficiency of $\eta = 1/10$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is

(A) 99 J

(B) 90 J

(C) 1 J

(D) 100 J

Answers & Solutions

Q.1 Answer:

373 K

Solution:

(d) Initially $\eta = \frac{T_1 - T_2}{T_1} \Rightarrow 0.5 = \frac{T_1 - (273+7)}{T_1} \Rightarrow \frac{1}{2} = \frac{T_1 - 280}{T_1} = T_1 = 560K$
Finally $\eta_1' = \frac{T_1' = T_2}{T_1'} \Rightarrow 0.7 = \frac{T_1' - (273+7)}{T_1'} \Rightarrow T_1' = 933K$ **∴.jncrease in temperature** $= 933 - 560 = 373K$

Q.2 Answer:

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487.5 g/min

Solution:

Coefficient of performance of the refrigerator, being ideal. is given by

$\omega = \frac{T_2}{T_1 - T_2} = \frac{273\ \text{K}}{(273+20\ \text{K}) - 273\ \text{K}} = \frac{273}{20}$

Now, if W be the average work done on the working substance her second, and Q_2 be the average heat extracted from a body at lower temperature, during that time, then

$\omega = \frac{Q_2}{W}$

or $Q_2 = \omega W = \frac{273}{20} \times 200\ \text{J/s} = 2730\ \text{J/s} = 650\text{cal/s}$

Now, if m be the mass of water frozen per minute, then, heat needed to be extracted will be

$Q = mL = m \times 80\text{cal/g}$

But the actual heat extracted per minute

$= 650 \frac{\text{cal}}{\text{s}} \times \frac{60\ \text{s}}{\text{min}} = 39 \times 10^3 \text{cal/min}$

$\Rightarrow 80\ \text{mcal/g} = 39 \times \text{J}^3\text{cal/min}$

or $m = 487.5\ \text{g/min}$

Q.3 Answer:

5.9

Solution:

Given. $T_1 = 273 + 30\ \text{K} = 303\ \text{K}$ **and**

$T_2 = 273 - 10\ \text{K} = 263\ \text{K}$

Coefficient of performance is given by

$\omega = \frac{T_2}{T_1 - T_2} = \frac{263\ \text{K}}{303\ \text{K} - 263\ \text{K}} = 5.9$

Q.4 Answer:

40 kcal

Solution:

Given $Q_1 =$ **50 kcal and** $\eta = 2\%$

Using $\eta = \frac{W'}{Q_1}$**, we have** $W = \eta \times Q_1$

i.e. work obtained per cycle

$W = 20\% \times 50\text{kcal} = 10\text{kcal} = 42\ \text{kJ}$

Since $Q_1 = Q_2 + W$

so, $Q_2 = Q - W$

i.e. heat rejected to the sink per cycle Q_2

$= 50\text{kcal} - 10\text{kcal} = 40\text{kcal}$

Q.5 Answer:

13.7 kW

Solution:

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Heat received (or produced by the burning of petrol) in one hour will be $= 2.4$ **kg/hour)**
$35\mathrm{MJ/kg}) = 95.2 \times 106$ J hour
∴ The rate at whicn heat is received $= \frac{85.2\times10^6 \text{ J}}{(3600 \text{ s})}$
$= 2.37 \times 104$ J s $= 23.7\mathrm{kW}$
Now, efficiency of the engine $\eta = \frac{\text{rate at which work is obtained}}{\text{rate at which heat is supplied}}$
$= \frac{10\mathrm{kW}}{23.7 \text{ kW}} = 42.2\%$
Evidently, the rate of heat rejection = rate at which heal is produced - rate at which work is obtained
$= 23.7 \text{ kW} - 10\mathrm{kW} = 13.7 \text{ kW}$

Q.6 Answer:

25 %

Solution:

Given, heat rejected per cycle, $Q_2 = 0.75Q_1$ **where** Q_1 **is the heat received per cycle.**
$\therefore W = Q_1 - Q_2 = Q_1 = 0.75Q_1 = 0.25Q$
∴ Efficiency, $\eta = \frac{W}{Q_1} = 0.25$ **or** 25%

Q.7 Answer:

62.5 %

Solution:

First, let us find the efficiency of the engine initially (which is not required here).
Using $\eta = 1 - \frac{T_2}{T_1}$
we have $\eta = 1 - \frac{(273-73)\mathrm{K}}{(273+127)\mathrm{K}} = 0.5$
When the temperature of the source is increased by $50^\circ\mathrm{C}$ **i.e.,**
$T_1 = 273 + 127 + 50 \text{ K} = 450 \text{ K}$
$\therefore \quad \eta_1 = 1 - \frac{200 \text{ K}}{450 \text{ K}} = 55.55\%$
When the temperature of the sink is decreased by $50^\circ\mathrm{C}$ **i.e,**
$T_2 = 273 - 73 - 50 \text{ K} = 150 \text{ K}$
$\therefore \quad \eta_2 = 1 - \frac{150 \text{ K}}{400 \text{ K}} = 62.5\%$
Evidently, $\eta_2 > \eta_1$ **i.e., reducing the temperature of the sink by** $50^\circ\mathrm{C}$ **proves to be more efficient.**

Q.8 Answer:

0.417 Kcal/s

Solution:

The efficiency of the heat engine is
$\eta = 1 - \frac{T_2}{T_1} = 1 - \left[\frac{273+27 \text{ K}}{273+427 \text{ K}}\right] = \frac{4}{7}$
But $\quad \eta = \frac{W}{Q_1}$
$\therefore \quad Q_1 = \frac{W}{\eta} = \frac{1.0 \text{ kW}}{4/7} = 1.75 \text{ kW} = 0.417\mathrm{kcal/s}$
Thus, the engine would require 417 cal of heat per second, to deliver the requisite amount of work.

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Q.9 Answer:

37 %

Solution:

Given, $\rho = \frac{V_3}{V_2} = 2$ and γ for a monatomic gas $= 5/3$.
Using: $\eta = 1 - \left(\frac{1}{\rho}\right)^{\gamma - 1}$, we have
Required efficiency, $\eta = 1 - \left(\frac{1}{2}\right)^{\frac{5}{3} - 1} = 1 - 0.63 = 0.37$ or 37%

Q.10 Answer:

100 K

Solution:

Let T_1 be the initial temperature of the source, then, using $\eta = 1 - \frac{T_2}{T_1}$,
we have, $\frac{40}{100} = 1 - \frac{(273 + 27\ \mathrm{K})}{T_1}$
or $\quad T_1 = 500\ \mathrm{K}$
For the efficiency to be 10% more, i.e., 50%,
let T_1', be the new temperature of the sink,
then, $\frac{50}{100} = 1 - \frac{(273 + 27\ \mathrm{K})}{T_1'}$
of $\quad T_1' = 600\ \mathrm{K}$
The required increase in the temperature of the source, $T_1' = T = 600\ \mathrm{K} - 500\ \mathrm{K} = 100\ \mathrm{K}$

Q.11 Answer:

26.81 %

Solution:

Ice point $= 0^\circ\mathrm{C} = 273\ \mathrm{K}$ and steam point $= 100^\circ\mathrm{C} = 373\ \mathrm{K}$.
That the higher temperature in a heat
We know that the higner temperature in a heat engine corresponds to the source (T_1) and the lower one to that of the sink (T_2).
Thermal efficiency, $\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{273}{373} = 0.2681$ or 26.81%

Q.12 Answer:

All reversible cycles have the same efficiency

Solution:

The correct choice is (a). The efficiency of a reversible cycle depends upon the temperature of the source and the sink which will be different for different cycles.

Q.13 Answer:

attain absolute zero temperature

Solution:

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The efficiency of Carnot engine is given

$\eta = 1 - \frac{T_2}{T_1}$

where T_1 is the absolute temperature of the source and T_2 that of the sink. To have 100% efficiency η must be equal to 1 which is possible if $T_2 = 0$ kelvin which is not attainable.

Q.14 Answer:

0.75

Solution:

Efficiency of the engine is

$\eta = 1 - \frac{T_2}{T_1}$

For an adiabatic process, $TV^{\gamma-1}+$ constant.

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$\Rightarrow \quad \frac{T_2}{T_1} = \left(\frac{V_2}{V_1}\right)^{\gamma-1}$$

For a diatomic gas $\gamma = \frac{7}{5}$. Hence

$$\frac{T_2}{T_1} = \left(\frac{32V}{V}\right)^{\left(\frac{7}{5}-1\right)} = (32)^{2/5} = 4$$

$$\therefore \quad \eta = 1 - \frac{1}{4} = \frac{3}{4} = 0.75$$

Q.15 Answer:

90 J

Solution:

The efficiency of a Carnot engine is given by

$$\eta = \frac{\text{Work done per cycle}}{\text{Heat energy supplied per cycle}}$$

$$= \frac{W}{Q_1} = 1 - \frac{T_2}{T_1}$$

where T_1 = temperature of the source and T_2 = temperature of the sink. Given $\eta = 10\% = 0.1$.

Hence

$0.1 = 1 - \frac{T_2}{T_1} \Rightarrow T_2 = 0.9 \text{ T}_1$

The performance of a refrigerator is given by

$$\beta = \frac{\text{Heat absorbed per cycle}}{\text{Work done per cycle}}$$

$$= \frac{Q_2}{W} = \frac{T_2}{T_1 - T_2} = \frac{0.9T_1}{T_1 - 0.9T_1} = 9$$

$\therefore Q_2 = 9W = 9 \times 10 = 90$ J

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